Exercise Sheet 25

Throughout, we use the Hermitian matrix

$$\Phi(x) = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}.$$

The pairing is $x \cdot y = x_0y_0 - x_1y_1 - x_2y_2 - x_3y_3$. For $g \in \mathrm{SL}(2,\mathbb{C})$, the map $\Lambda(g)$ is defined by

$$\Phi(\Lambda(g)x) = g\Phi(x)g^{*}.$$

Each map $\Lambda(g)$ is linear and preserves the pairing, and Λ is multiplicative. The group $G = \mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$ has product $(a,g)(a',g') = (a + \Lambda(g)a', gg')$. Exercises 25.1–25.6 are core. Exercise 25.7 is an optional extension.

Exercise 25.1 (Central element and spin parity). Let $z = (0,-I) \in G$.

(a) Show that $\Lambda(-I) = 1$, that z is central, and that $z^2 = (0,I)$.

(b) Let U be a unitary representation of G on a Hilbert space $\mathcal{H}$ whose only closed invariant subspaces are 0 and $\mathcal{H}$. Show that $U(z) = \varepsilon \cdot 1$ with $\varepsilon = \pm 1$.

(c) Let $m > 0$. For p with $p \cdot p = m^2$ and $p_0 > 0$, let h_p be the positive square root of $\Phi(p)/m$, and let $W(g,p) = h_p^{-1} g h_{\Lambda(g)^{-1}p}$. Let D_n act on homogeneous polynomials of degree n in $w \in \mathbb{C}^2$ by $(D_n(u)f)(w) = f(u^{-1}w)$. Consider the model

$$(U(a,g)\psi)(p) = e^{-ip\cdot a} D_n(W(g,p))\psi(\Lambda(g)^{-1}p).$$

Compute $U(z)$ in this model.

Exercise 25.2 (Characters of $\mathbb{R}^4$).

(a) Let $\varphi\colon \mathbb{R} \to \mathrm{U}(1)$ be a continuous homomorphism. Show that there is $\delta > 0$ with $c = \int_0^\delta \varphi(s)\,ds \neq 0$. Deduce that φ is differentiable and $\varphi(t) = e^{\kappa t}$ with $\kappa \in i\mathbb{R}$.

(b) Deduce that every continuous homomorphism $\chi\colon \mathbb{R}^4 \to \mathrm{U}(1)$ equals $\chi_p(a) = e^{-ip\cdot a}$ for a unique $p \in \mathbb{R}^4$.

(c) Show that $\chi_p(\Lambda(g)^{-1}a) = \chi_{\Lambda(g)p}(a)$.

Exercise 25.3 (Explicit Lorentz matrices). Use coordinates (p_0,p_1,p_2,p_3). Write $A = p_0 + p_3$, $D = p_0 - p_3$ and $B = p_1 + ip_2$, so that

$$\Phi(p) = \begin{pmatrix} A & \bar{B} \\ B & D \end{pmatrix}.$$

(a) Let $\eta \in \mathbb{R}$, $s = e^{\eta/2}$ and $d_s = \mathrm{diag}(s, s^{-1})$. Compute the 4×4 matrix of $\Lambda(d_s)$.

(b) Compute the matrix of $\Lambda(r(e^{-i\theta/2}))$, where $r(\lambda) = \mathrm{diag}(\lambda, \bar{\lambda})$.

(c) For real β let

$$n(\beta) = \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}.$$

Compute the matrix of $\Lambda(n(\beta))$. Check that it fixes $k = (1,0,0,1)$. Compute its value on $(1,0,0,-1)$ and check that the image is lightlike.

Exercise 25.4 (Cartan decomposition and the invariant measure). Let $m > 0$, $\omega(\vec{p}) = \sqrt{m^2 + |\vec{p}|^2}$, and $X_m^+ = \{p \cdot p = m^2,\ p_0 > 0\}$. Define $\int f\, d\mu_m = \int_{\mathbb{R}^3} f(\omega(\vec{p}), \vec{p})\, d^3\vec{p}/(2\omega(\vec{p}))$ for nonnegative Borel f on X_m^+. Let $k_1 = (1,0,0,0)$ and $d_s = \operatorname{diag}(s, s^{-1})$. You may use the following fact from the orbit proof: for each $p \in X_1^+$ there are $u \in \mathrm{SU}(2)$ and $s > 0$ with $\Lambda(u d_s) k_1 = p$.

(a) Show that every $g \in \mathrm{SL}(2,\mathbb{C})$ factors as $g = u\, d_s\, v$ with $u, v \in \mathrm{SU}(2)$ and $s > 0$.

(b) For $u \in \mathrm{SU}(2)$, show that $\Lambda(u)$ fixes p_0 and acts on $\vec{p}$ by an orthogonal matrix. Deduce that $\int f(\Lambda(u)^{-1} p)\, d\mu_m = \int f\, d\mu_m$.

(c) Let $s = e^{\eta/2}$. Let $F(\vec{p})$ be the spatial part of $\Lambda(d_s)^{-1}(\omega(\vec{p}), \vec{p})$. Show that the Jacobian determinant of F is $\omega(F(\vec{p}))/\omega(\vec{p})$, and deduce invariance under $\Lambda(d_s)$.

(d) Conclude that μ_m is invariant under every $\Lambda(g)$.

Exercise 25.5 (The massive model is induced). Let $m > 0$, $k_m = (m,0,0,0)$ and $\chi_p(a) = e^{-ip\cdot a}$. Let $G_0 = \{(a,u) : a \in \mathbb{R}^4,\ u \in \mathrm{SU}(2)\}$. Let D_n be a unitary representation of $\mathrm{SU}(2)$ on V_n. Define h_p and $W(g,p)$ as in Exercise 25.1, and the model U by the same formula.

(a) Show that G_0 is a closed subgroup and that $\sigma(a,u) = \chi_{k_m}(a) D_n(u)$ is a representation of G_0.

(b) Let $x = (a,g) \in G$ and $p = \Lambda(g) k_m$. Find $y \in G_0$ with $x = (0, h_p)\, y$.

(c) Let $\mathcal{F}$ be the space of continuous $f : G \to V_n$ with $f(xy) = \sigma(y)^{-1} f(x)$ for $y \in G_0$. Let $(\lambda(x) f)(x') = f(x^{-1}x')$, and put $\psi_f(p) = f(0, h_p)$. Show that $\psi_{\lambda(x)f} = U(x)\psi_f$.

(d) Express $f(a,g)$ through ψ_f. Deduce that $f \mapsto \psi_f$ is injective and that $\|f(a,g)\| = \|\psi_f(\Lambda(g)k_m)\|$.

(e) Take $m = 1$, $g = \operatorname{diag}(2, \tfrac{1}{2})$ and $h = \begin{pmatrix} 2 & \tfrac{3}{2} \\ \tfrac{3}{2} & \tfrac{13}{8} \end{pmatrix}$. Let q be the momentum with $\Phi(q) = h^2$, so that $h_q = h$, and put $p = \Lambda(g)q$. Show that $h_p = \frac{1}{1360}\begin{pmatrix} 6656 & 1392 \\ 1392 & 569 \end{pmatrix}$. Compute $W(g,p)$, and show that it equals $\exp(\theta X_2)$ with $X_2 = -\frac{i}{2}\sigma_2$ and $\cos(\theta/2) = \frac{77}{85}$. Show also that $W(g,p) = I$ when g and h_q are both diagonal.

Exercise 25.6 (Spacelike and massless little groups). Let $\sigma_3 = \operatorname{diag}(1,-1)$. For $b \in \mathbb{C}$ and $|\lambda| = 1$, let

$$n(b) = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, \qquad r(\lambda) = \operatorname{diag}(\lambda, \bar{\lambda}).$$

Let $H_k = \{n(b) r(\lambda)\}$.

(a) Show that $g \in \mathrm{SL}(2,\mathbb{C})$ satisfies $g\sigma_3 g^* = \sigma_3$ exactly when there are $\alpha, \beta \in \mathbb{C}$ with

$$g = \begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix}, \qquad |\alpha|^2 - |\beta|^2 = 1.$$

Show that this group is not compact.

(b) Show that the defining action of H_k on $\mathbb{C}^2$ preserves no inner product.

(c) For $\nu \in \mathbb{Z}$, show that $\sigma_\nu(n(b)r(\lambda)) = \lambda^\nu$ is a continuous homomorphism. Compute $\sigma_\nu(-I)$. Let U be the representation induced from $\chi_k \otimes \sigma_\nu$, realised by the equivariant-function action of Exercise 25.5(c), now with subgroup $\mathbb{R}^4 \rtimes H_k$ and $k = (1,0,0,1)$. Compute $U(z)$ for $z = (0, -I)$.

Exercise 25.7 (Optional extension: a continuous-spin model). Fix $\rho > 0$ and $\varepsilon \in \{0, 1\}$. Let $\mathcal{K}$ be the Hilbert space of 2π-periodic square-integrable functions, with norm $\|f\|^2 = \frac{1}{2\pi}\int_0^{2\pi} |f(\alpha)|^2\, d\alpha$. For $b \in \mathbb{C}$ and $\theta \in \mathbb{R}$ define

$$(\sigma(n(b))f)(\alpha) = e^{-i\rho\,\mathrm{Re}(e^{-i\alpha}b)} f(\alpha), \qquad (\sigma(r(e^{i\theta}))f)(\alpha) = e^{i\varepsilon\theta} f(\alpha - 2\theta),$$

and $\sigma(n(b)r(\lambda)) = \sigma(n(b))\sigma(r(\lambda))$. Use $n(b)$, $r(\lambda)$ and H_k as in Exercise 25.6.

(a) Show that $\sigma(r(\lambda))$ is well defined and that $\sigma(r(\lambda))\sigma(n(b))\sigma(r(\lambda))^{-1} = \sigma(n(\lambda^2 b))$. Deduce that σ is a unitary representation of H_k. You may assume continuity.

(b) Show that σ is nontrivial on $N = \{n(b)\}$.

(c) Compute $\sigma(r(\lambda))$ on $e_j(\alpha) = e^{ij\alpha}$, and find all ν for which λ^ν occurs.

(d) Prove that σ is irreducible. Let P be the orthogonal projection onto a closed invariant subspace. First use the distinct T-weights to show that P is diagonal in the basis e_j. Then differentiate the bounded multiplication operators $\sigma(n(t))$ and $\sigma(n(it))$ at $t = 0$.